

Pietro Costanzi Fantini

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in Pietro Costanzi Fantini 🌐 Nabster101

Education

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|------------|---|-----------------------|
| EXC | University of Porto , Computer Science (Erasmus+ Exchange) <ul style="list-style-type: none"> • Awarded Erasmus+ Scholarship • Coursework: Introduction to Data Science, Interactive Graphics System, Software Testing Verification and Validation, Cryptography | Sept 2025 – Feb 2026 |
| MSc | Sapienza University of Rome , M.Sc. in Computer Science <ul style="list-style-type: none"> • Average: 28/30 • Coursework: Security, Data Science, Computer Systems | Sept 2024 – Present |
| BSc | Sapienza University of Rome , B.Sc. in Computer Engineering <ul style="list-style-type: none"> • Thesis: Web vulnerabilities and cyberattacks: An investigation into exploitation techniques • Coursework: Mathematics, Computer Science, Physics, Automation | Sept 2021 – Sept 2024 |

Work Experience

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| Toyota Racing , Software Developer - Internship <ul style="list-style-type: none"> • Created a live aggregation system for telemetry data using Apache Flink and Apache Kafka. • Setup an ingestion pipeline for Amazon Web Services data. | Mar 2026 – Present |
| Fibernet , Software Developer - Internship <ul style="list-style-type: none"> • Created a temperature measurement system to monitor the behavior of fiber optic cables at various temperatures. • Developed functional and technical documentation for the application system. | Nov 2019 – Mar 2020 |

Projects

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| Adaptive Cache Occupancy Fingerprinting <ul style="list-style-type: none"> • Researched adaptive cache-based website fingerprinting that selects device-specific ML models, boosting classification accuracy from 50% to 96%. • Tools Used: Python, Tensorflow, Keras | May 2025 – Jun 2025
Link  |
| Comparative Analysis of Fingerprint Biometric Systems <ul style="list-style-type: none"> • Developed a complete fingerprint recognition system to compare enhancement algorithms across multiple datasets, implementing feature extraction pipelines and computing performance metrics with ROC curves, EER, and decidability scores. • Tools Used: Tools Used: Python, OpenCV, NumPy, Matplotlib | Feb 2025 – Mar 2025
Link  |
| Kubernetes Orchestration on AWS with HPA <ul style="list-style-type: none"> • A self-managed Kubernetes cluster on AWS runs a scalable web app to evaluate Horizontal Pod Autoscaler behavior under varying traffic using Apache JMeter. • Tools Used: Docker, Kubernetes, Containerd, AWS, Flannel, JMeter | Aug 2025 – Sept 2025
Link  |
| AES vs SHA3 using GPU acceleration <ul style="list-style-type: none"> • An overview of AES and SHA-3 focusing on algorithm structure, parallelizable modes, and high-performance implementations using AES-NI on CPUs and CUDA techniques on GPUs. | Jun 2025 – Jul 2025
Link  |

Dunkzone

Apr 2024 – Jun 2024

[Link](#)

- Developed an online utility portal with the use of the official NBA API where users can keep track of all upcoming and past matches of their favorite NBA team, along with statistics and other features.
- **Tools Used:** JS, Python, React

File System FAT

Feb 2024 – Mar 2024

[Link](#)

- Developed a fully fledged FAT system in C with the added feature of being persistent. It includes useful features such as a write-on-file function, a read-from-file, a make-directory command and others.
- **Tools Used:** C

Technologies

Languages: C++, C, Java, Python, SQL, JavaScript/TypeScript

Data Science: Python (Pandas, NumPy, scikit-learn), TensorFlow, Keras, Matplotlib, OpenCV

Cloud Computing: Kubernetes, Containerd, AWS, Docker

Graphic Systems: ThreeJS, CUDA

Web Development: Flask, React, Next.js, Node.js, HTML/CSS, RESTful APIs

Tools & Platforms: Git/GitHub, Linux, LaTeX, Jupyter

Languages

Italian: Native

English: Cambridge C2

Volunteering

AGESCI (Italian Catholic Guides and Scouts Association)

Oct 2009 – Present

CRI (Croce Rossa Italiana - Italian Red Cross) volunteer

Apr 2024 – Present

Boy Scouts of America (BSA)

Sept 2017 – Jul 2019